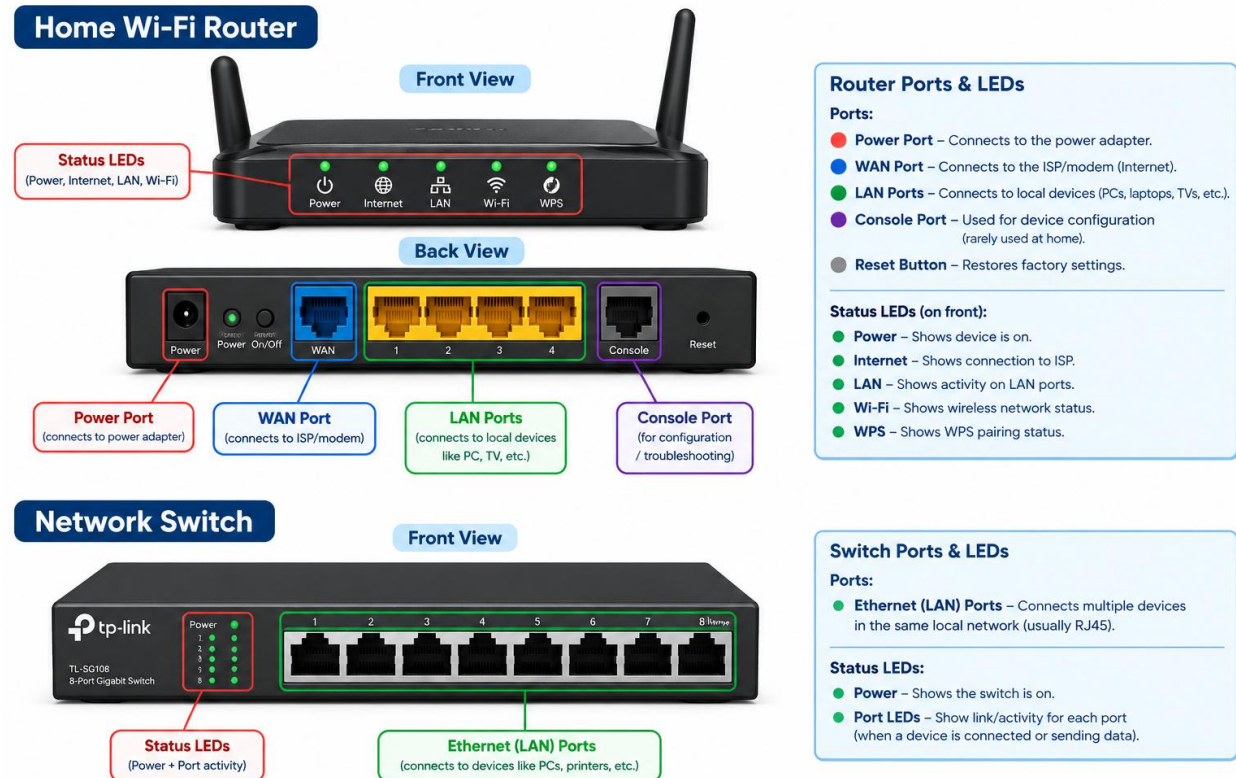


ASSIGNMENT 3.2

1. Draw a labelled diagram of a typical home Wi-Fi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.

Answer:-



2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

Answer:-

Difference	Router	Switch
1. Main Function	Connects different networks, such as a LAN to the Internet.	Connects multiple devices within the same LAN.
2. Address Used	Uses IP addresses to route packets.	Uses MAC addresses to forward frames.
3. Traffic Handling	Directs traffic between different networks .	Forwards traffic within the same network to the correct device.

3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

Answer:-

Common Uses of Network Cables

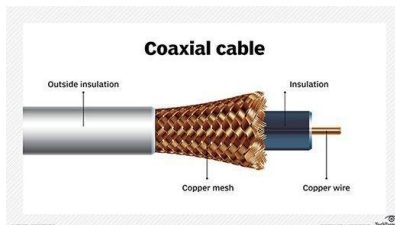
1. Ethernet Cable (RJ-45):

Ethernet cables connect computers, laptops, printers, routers, switches, and other network devices. They are mainly used in LANs for reliable wired internet connections in homes, offices, schools, and labs.



2. Coaxial Cable:

Coaxial cables are used for cable TV, broadband internet, CCTV, and antenna connections. They provide protection against signal interference.



3. Fiber Optic Cable:

Fiber optic cables are used for high-speed internet, telecommunications, data centers, and long-distance network connections. They transmit data using light and provide high speed with low signal loss.



4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

Answer:-

Small Gaming Café Network

Item	Scenario / Explanation
ISP Line	The ISP internet line is connected first to the router .
Router	The router receives the internet connection, manages network traffic, and provides IP addresses to the devices.
Switch	The router is connected to an 8-port or 16-port switch using an Ethernet cable.
8 Gaming PCs	Each of the 8 gaming PCs is connected to a separate port on the switch using Ethernet cables.
Why Router First?	The router is required to connect the café's local network to the internet and manage the connection.
Why Switch?	The switch provides multiple Ethernet ports so all 8 gaming PCs can connect to the same network efficiently.
Why Wired Connection?	Ethernet connections provide stable and low-latency performance, which is useful for multiplayer gaming and smooth IPL streaming.
Final Connection	ISP → Router → Switch → 8 Gaming PCs